

INSTITUTE OF ACTUARIES OF INDIA
EXAMINATIONS

May 2026

CB2-Business Economics

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

Q.1) During a prolonged period of low interest rates, traditional commercial banks observed aggressively expanding fintech firms generating record short-term profits by issuing uncollateralized loans to early-stage artificial intelligence (AI) startups. Despite internal macroeconomic risk models explicitly flagging these tech sector loans as highly speculative and vulnerable to supply chain shocks, traditional bank executives aggressively entered the AI lending market and dismantled their own underwriting safeguards. They did this specifically to prevent losing market share to the fintech firms. When the AI sector experienced a sudden contraction in revenue, a cascade of defaults triggered a widespread banking panic.

Which of the following behavioural economics concepts best explains the actions of the traditional commercial bank executives leading up to this crisis?

- A. Disaster myopia, because the commercial banks systematically underestimated the probability of a low-frequency shock due to a prolonged period of macroeconomic stability.
- B. Moral hazard, because the commercial bank executives assumed that the central bank would act as a lender of last resort to bail out systemic failures in the tech sector.
- C. Herding behaviour, because the banks intentionally bypassed their own independent risk assessments to blindly mimic the profitable actions of competing firms.
- D. Adverse selection, because the dismantling of underwriting standards inherently attracted the most financially fragile and risky AI startups in the market.
- E. Irrational exuberance, because the bank executives genuinely believed the fundamental market value of AI startups would perpetually increase without correction.

[2 marks]

Q.2) When evaluating the strategic interactions of firms in an oligopoly using game theory, which of the following theoretical statements regarding dominant strategies, Nash equilibria, and game structures is completely accurate?

- A. A Nash equilibrium can only be achieved in a simultaneous single-move game if both interacting firms possess a strict dominant strategy.
- B. In a classic Prisoner's Dilemma, the Nash equilibrium is always reached when firms play their dominant strategies, though this outcome represents a sub-optimal position compared to potential collusion.
- C. Converting a simultaneous single-move game into a sequential-move game will inherently force the second-moving firm to abandon its dominant strategy in response to a credible threat.
- D. A firm adopting a maximax strategy will actively seek the Nash equilibrium outcome, assuming the competing firm will also make decisions to maximize joint industry profits.
- E. If a firm possesses a dominant strategy in a simultaneous game, it will automatically gain a first-mover advantage if the rules are changed to a sequential-move game.

[2 marks]

Q.3) Rohan, a cybersecurity analyst, derives utility (job satisfaction from completed audits) using two inputs: Hours of AI Processing Time (A) and Hours of Manual Code Review (M). His firm's protocol strictly dictates that completing a single valid audit report requires exactly 2

hours of AI Processing Time and 5 hours of Manual Code Review. Any surplus of one input without the exact corresponding proportion of the other provides zero additional utility to Rohan.

Assume A is plotted on the horizontal axis and M is plotted on the vertical axis. Based on Rohan's preferences, what is the geometric shape of his indifference curves, and what is the Marginal Rate of Substitution of AI Processing Time for Manual Review ($MRS_{A, M}$) if his current allocated bundle is 4 hours of A and 5 hours of M?

- A. Right-angled (L-shaped); $MRS_{A, M} = \infty$
- B. Right-angled (L-shaped) $MRS_{A, M} = 0$
- C. Linear (straight-line); $MRS_{A, M} = 0.4$
- D. Convex to the origin; $MRS_{A, M}$ is diminishing
- E. Right-angled (L-shaped); $MRS_{A, M} = 2.5$

[2 marks]

Q.4) AlphaRide is a ride-hailing firm in a competitive city market. Output Q is rides per week (in thousands). Costs (₹000) are: $TC = 150 + 10Q + 0.5Q^2$. A new subsidised metro service reduces demand, so the market price falls to ₹15 per ride. Hence $TR = 15Q$. In the short run, what should AlphaRide do to minimise losses?

- A. Produce $Q = 5$ because $MR = MC$ and it earns normal profit.
- B. Produce $Q = 5$ because $P > AVC$, so it covers variable cost and contributes to fixed cost
- C. Shut down because $P < \min(AVC)$
- D. Shut down because $TR < TC$ (economic loss)
- E. Produce $Q = 30$ to maximise revenue and offset losses

[2 marks]

Q.5) In response to a sudden surge in demand for commercial cloud storage, the global data infrastructure firm "NexaCloud" increased the price of its premium 10-terabyte enterprise plan from \$400 to \$500 per month. Consequently, NexaCloud's data centers increased their monthly provision of premium server space from 20,000 terabytes to 30,000 terabytes.

Simultaneously, an independent industry analysis observed that the average income of mid-sized tech startups (NexaCloud's primary customer base) grew from \$1.2 million to \$1.6 million annually. As a result of this income growth, the total market demand for commercial cloud storage increased from 50,000 terabytes to 60,000 terabytes per month.

Assuming ceteris paribus conditions for each calculation, what are the values of the price elasticity of supply (using the original quantities method) and the income elasticity of demand (using the average quantities/midpoint method), respectively?

- A. Price elasticity of supply = 0.50; Income elasticity of demand = 1.50
- B. Price elasticity of supply = 2.00; Income elasticity of demand = 0.64
- C. Price elasticity of supply = 2.00; Income elasticity of demand = 0.55

D. Price elasticity of supply = 2.50; Income elasticity of demand = 0.64

E. Price elasticity of supply = 1.25; Income elasticity of demand = 0.55

[2 marks]

Q.6) In the nation of Zephyria, the central bank unexpectedly maintains a highly accommodative monetary policy, causing the annual inflation rate to surge to 9%, significantly above the historically anticipated 2%. "AeroLogistics", a mid-sized national freight forwarding company, is assessing the impact of this macroeconomic shock on its operations.

The firm currently holds a substantial amount of long-term, fixed-rate debt from a recent warehouse and fleet expansion. Furthermore, unlike traditional competitors who print physical quarterly rate cards, AeroLogistics utilizes an automated, cloud-based digital platform that updates client shipping tariffs in real-time.

Which of the following statements most accurately evaluates the economic costs and impacts of this unexpected inflation on AeroLogistics' business activity?

- A. The real value of their fixed-rate debt will fall, transferring wealth from their lenders to AeroLogistics, but the macroeconomic uncertainty created by the inflation surge is likely to depress their future long-term investment plans.
- B. The real value of their fixed-rate debt will rise, severely restricting their operational cash flow, while the firm will simultaneously suffer from exorbitant menu costs as they continuously update prices.
- C. The redistribution of wealth caused by the inflation will heavily favour their creditors, while the entirely predictable nature of the price level increases will stimulate AeroLogistics to accelerate capital expenditures.
- D. AeroLogistics will experience a drastic reduction in shoe-leather costs, as the company will rationally choose to hold substantially larger idle cash balances to finance their increasingly expensive daily operations.
- E. The unanticipated demand-pull inflation will automatically trigger a deflationary spiral in the broader logistics sector, allowing AeroLogistics to permanently lower nominal wages without affecting real productivity.

[2 marks]

Q.7) Consider the artisanal soy candle industry, which is perfectly competitive. There are thousands of identical sellers, no barriers to entry or exit, perfect information, and the soy candles are homogeneous. Currently, the industry is in long-run equilibrium. An unexpected viral trend dramatically increases demand for soy candles. Because soy wax is derived from soybeans, expanding industry output drives up the cost of agricultural land used for cultivation. What will be the ultimate long-run outcome for the industry?

- A. The industry will return to its original price and output levels, with firms earning normal profit.
- B. The industry will settle at a higher price and higher total output, with existing firms earning supernormal profit.
- C. The industry will settle at a higher price and higher total output, with firms earning normal profit.

- D. The industry will settle at the original price but with higher total output, with firms earning normal profit.
- E. The industry will settle at a lower price and higher total output, with firms earning normal profit.

[2 marks]

Q.8) According to the monetarist (expectations-augmented) Phillips curve framework, what happens to the relationship between inflation and unemployment if a central bank attempts to permanently hold unemployment below its natural rate using continuous expansionary monetary policy?

- A. The economy moves leftward along a stable, downward-sloping long-run Phillips curve, achieving a permanently lower unemployment rate at the cost of a higher, but constant, inflation rate.
- B. The short-run Phillips curve shifts downwards as workers suffer from money illusion, restoring the natural rate of unemployment while permanently reducing the rate of inflation.
- C. The short-run inverse relationship breaks down as workers revise their inflation expectations upwards, causing the short-run Phillips curve to shift vertically upwards and resulting in accelerating inflation at the natural rate of unemployment.
- D. The long-run Phillips curve shifts to the left because the continuous monetary stimulus permanently increases the productive capacity of the economy, creating a new, lower natural rate of unemployment.
- E. The inverse relationship remains perfectly stable in both the short run and long run, meaning inflation will only accelerate if the government simultaneously implements expansionary fiscal policy.

[2 marks]

Q.9) The Central Bank of Meridian (CBM) faces soaring retail inflation driven by a massive surge in household borrowing. Consumers are taking out personal bank loans to invest in the country's booming domestic green-tech sector, which has been experiencing unprecedented exponential growth. In an attempt to curb this unchecked credit expansion, the CBM aggressively raises its primary policy interest rate by 200 basis points.

Despite this restrictive monetary policy, retail credit growth continues to expand rapidly over the next two quarters.

Which of the following best explains why the CBM's use of interest rates has failed to control credit in this specific scenario?

- A. Commercial retail banks have perfectly and immediately transmitted the 200-basis-point policy hike to their variable retail lending rates.
- B. The higher interest rates have significantly increased the opportunity cost of consumption, leading households to prioritize saving over spending.
- C. The anticipated capital gains from green-tech investments heavily outweigh the increased borrowing costs, making household credit demand highly interest-inelastic.
- D. The CBM's policy rate hike primarily anchors long-term corporate bond yields, having almost no knock-on effect on the short-term retail loans utilized by households.

- E. The aggressive increase in interest rates has pushed the economy into a liquidity trap, rendering the central bank incapable of altering the broad money supply.

[2 marks]

Q.10) The government of the fictional nation of Novaria has recently announced a new policy offering substantial "R&D Expenditure Credits" to domestic pharmaceutical companies engaged in researching long-term treatments for degenerative neurological conditions. Prior to this intervention, research in this specific medical field had stagnated, with companies primarily focusing on minor, easily marketable iterations of existing generic drugs.

Which of the following best describes the primary economic rationale for why the Novarian government chose this specific intervention (R&D Expenditure Credits) to address the market failure in this scenario?

- A. To directly prevent pharmaceutical firms from acting as "free riders" who copy the research findings of their competitors without incurring any initial costs.
- B. To eliminate the monopolistic structures within the Novarian pharmaceutical industry, thereby forcing firms to innovate to survive.
- C. To reduce the high financial costs and mitigate the substantial uncertainty associated with long-term, high-risk research that firms were previously unwilling to undertake.
- D. To centralize research efforts under government control, thereby eliminating the inefficiencies caused by multiple private firms duplicating the same research.
- E. To implement stricter competition policies and regulatory standards, ensuring that all new treatments meet high safety and compatibility requirements.

[2 marks]

Q.11) Vesperia's robotics sector has driven a four-year economic boom. Currently, corporate borrowing is at record highs to fund capital purchases, while unsold finished robotics (stocks) have suddenly surged relative to sales. Based on business cycle theory, which statement best evaluates this scenario?

- A. Driven by lags and optimism, the boom's persistence is ending; involuntary stock accumulation and peak debt signal an impending upper turning point.
- B. The stock increase is a deliberate supply-side response to the multiplier effect, guaranteeing continued economic expansion despite high debt.
- C. Record borrowing indicates the economy just passed the lower turning point (trough), as firms maximize debt to replace depreciated capital.
- D. Stock accumulation and high debt act as an economic "floor," preventing the economy from entering a recessionary slump.
- E. The boom persists because the economy hit its full capacity "ceiling," forcing rapid stock accumulation to sustain growth.

[2 marks]

Q.12) The nation of Veridia is experiencing a deep depression with 14% unemployment and widespread idle manufacturing capacity. Labor unions have locked in strict contracts

prohibiting nominal wage cuts, while long-term unemployed workers are losing technical skills (hysteresis). To stimulate the economy, the Central Bank aggressively buys government bonds.

Based strictly on the Keynesian AD-AS model, which statement best explains the shapes of Veridia's AS curves and the impact of this policy?

- A. The short-run aggregate supply (SRAS) curve is horizontal because idle capacity and rigid wages allow output growth without inflation; the long-run aggregate supply (LRAS) curve is non-vertical because hysteresis permanently impairs long-run productive capacity.
- B. The SRAS curve is upward-sloping due to rising marginal hiring costs; the LRAS curve is vertical because the economy will naturally self-correct regardless of the stimulus.
- C. The SRAS curve is vertical due to rigid nominal wage contracts preventing market clearing; the LRAS curve is horizontal because monetary policy can infinitely expand output.
- D. The SRAS curve is horizontal due to spare capacity; the LRAS curve is vertical because money neutrality ensures the policy only affects long-run prices.
- E. The stimulus will directly shift the SRAS curve downward; the LRAS curve remains non-vertical because nominal wages will immediately adjust downwards to clear the market.

[2 marks]

Q.13) To stimulate the economy, the Reserve Bank of India buys ₹750 billion in government securities, successfully lowering commercial lending rates by 150 basis points. However, domestic electric vehicle (EV) and semiconductor firms freeze planned factory expansions, citing deep pessimism over future consumer demand. Consequently, national income remains stagnant.

Which of the following best explains why this expansionary monetary policy failed?

- A. A Stage 1 breakdown (money supply–interest rate link) due to a liquidity trap where speculative money demand is perfectly interest-elastic.
- B. A Stage 2 breakdown (interest rate–investment link) because pessimistic business expectations have shifted investment demand left, making it unresponsive to lower rates.
- C. A Stage 3 breakdown (investment–national income link) because a high marginal propensity to withdraw severely restricted the Keynesian multiplier.
- D. A Stage 2 breakdown (interest rate–investment link) because the transactions demand for money is perfectly inelastic.
- E. A breakdown caused by financial crowding out, as the central bank's bond purchases absorbed available market funds.

[2 marks]

Q.14) The Republic of Meridian recently granted its central bank full political independence, legally mandating a strict inflation target of 2.0% as its singular primary objective. Six months into this new monetary regime, a severe global geopolitical conflict causes a massive, sustained spike in imported crude oil prices, heavily impacting Meridian's manufacturing and transport sectors.

Assuming the central bank strictly adheres to its new inflation-targeting framework, what is the most likely short-term monetary policy action and its immediate macroeconomic consequence for Meridian's business cycle and labour market?

- A. The central bank will raise interest rates to suppress the cost-push inflation, leading to an amplified contraction in real GDP and a sharp rise in short-term unemployment.
- B. The central bank will lower interest rates to mitigate the adverse supply shock, prioritizing the stabilization of the business cycle and labour markets over the 2.0% inflation target.
- C. The central bank's newfound independent credibility will immediately anchor public expectations, neutralizing the energy price shock and maintaining 2.0% inflation with no change to cyclical unemployment.
- D. The central bank will mandate price ceilings on domestic energy markets, ensuring the inflation target is successfully met while avoiding a domestic recession.
- E. The central bank will expand the money supply to boost aggregate demand, which will simultaneously bring inflation back down to the 2.0% target and restore short-term economic growth.

[2 marks]

Q.15) CloudServe is a well-established dominant firm operating in the regional corporate cloud-hosting market. CloudServe currently enjoys significant economies of scale and faces a downward-sloping demand curve.

A newly formed tech start-up, NovaHost, is considering entering this market to compete directly with CloudServe. In response, the management of CloudServe has decided to adopt a limit pricing strategy rather than maximizing short-run profits.

Explain, using cost and revenue curves, how CloudServe will implement a limit pricing strategy to deter NovaHost from entering the market, and discuss the specific conditions required for this strategy to be successful in this context.

Graphs are not required to be drawn in the answer.

[4 marks]

Q.16) Between January and December 2025, the Reserve Bank of India (RBI) implemented a cumulative 125 basis-point cut to its repo rate, lowering it to 5.25%. In tandem, the RBI announced a phased 100-basis-point reduction in the Cash Reserve Ratio (CRR) from 4% to 3%. This CRR reduction was explicitly intended to release roughly ₹2.5 lakh crore of primary liquidity into the banking system to support economic growth.

Explain the mechanisms through which these two policy actions are intended to increase the broad money supply, and discuss why the ultimate growth in broad money might remain weaker than the RBI anticipates.

[4 marks]

Q.17) ZephyrCloud is a software-as-a-service (SaaS) provider offering specialized cloud architecture optimization tools. As a price maker, it successfully prevents resale and engages in third-degree price discrimination across two distinct user segments: Corporate Enterprises (Market 1) and Independent Developers (Market 2).

The inverse demand functions for the two markets are given by:

- Corporate Enterprises:

$$P_1 = 200 - 5Q_1$$

- Independent Developers:

$$P_2 = 160 - 5Q_2$$

where

P is the monthly subscription price in dollars, and Q is the number of subscriptions sold per month in thousands.

ZephyrCloud develops and hosts its software centrally, meaning its total monthly cost TC depends on the combined total output Q where $Q = Q_1 + Q_2$

The firm's monthly total cost function, measured in thousands of dollars, is:
 $TC = 100 + 20Q + 2.5Q^2$

- i) Determine the Marginal Revenue functions for both markets individually, and the firm's Marginal Cost function expressed in terms of Q_1 and Q_2 . [1.5]
- ii) Using the profit-maximising conditions for a price-discriminating monopoly, calculate the exact number of subscriptions ZephyrCloud should sell in each market Q_1 and Q_2 . [2.5]
- iii) Calculate the profit-maximising price to be charged in each market, and the resulting total maximum monthly profit. [1.5]
- iv) Calculate the point price elasticity of demand (PED) in the Corporate Enterprises market at the profit-maximising price and quantity. [0.5]

[6 marks]

Q.18) Neha is a freelance software developer. Due to her university schedule, she is strictly limited to working a maximum of 30 hours per week. She divides her limited time between two mutually exclusive tasks: managing databases (Good X) and designing web pages (Good Y). The table below outlines her maximum production possibilities for a given week.

Production Alternative	Databases Managed (X)	Web Pages Designed (Y)
A	0	50
B	1	46
C	2	38
D	3	24
E	4	0

- i) Identify the principal source of scarcity in Neha's situation and explain it forces her to make an economic choice. [1]
- ii)
 - a) Calculate the opportunity cost of increasing her output from 1 database to 2 databases.

- b) Describe the impact on Neha's opportunity cost as her production of databases increases from 0 to 4.

[1]

Neha charges Rs. 5000 per database and Rs. 500 per webpage designed.

- iii) Assuming she wants to act as a rational economic agent, identify the production alternative that maximises her revenue. Consider only the alternatives where both tasks are non-zero.

[2]

[4 marks]

Q.19) SkyRoute Airways is a major commercial airline operating out of Mumbai. The board of directors is executing a massive five-year expansion strategy to double their fleet of aircraft, acquire new maintenance hangars at major hubs, and overhaul their pilot and cabin crew training programs.

The Chief Financial Officer (CFO) has projected that this expansion will initially lower the average cost per passenger but has warned that expanding beyond a certain point will inevitably lead to higher average costs.

- i) Explain four ways SkyRoute Airways could achieve the lower average costs (economies of scale or scope) projected by the CFO during the initial expansion phase.

[2]

- ii) Explain four specific factors that could lead to the higher average costs (diseconomies of scale) warned about by the CFO if the expansion goes too far.

[2]

[4 marks]

Q.20)

- i) Explain the primary supply-side factors that determine the potential rate of economic growth of a nation.

[3]

- ii) Discuss the institutional and structural barriers that cause certain nations to experience persistently lower economic growth rates than others.

[3]

[6 marks]

Q.21) The domestic market for a specialized electric vehicle (EV) battery has the following daily demand and supply functions:

$$Q_d = 12000 - 15P$$

$$Q_s = -3000 + 10P$$

where

P is the price in thousands of Rupees (INR '000) and

Q is the quantity of batteries per day.

- i) Calculate the initial free-market equilibrium price and quantity.

[1]

To promote domestic manufacturing, the government implements two simultaneous policy changes:

1. An import tariff on raw materials which increases the marginal cost of production, reducing the quantity supplied by 500 units at every price level.
 2. A consumer subsidy which increases the quantity demanded by 1,000 units at every price level.
- ii) State the new demand and supply equations, and calculate the new equilibrium price and quantity. [2]
- iii) Calculate the Point Price Elasticity of Demand (PED) at the new equilibrium point. State, with a brief reason, whether demand is elastic or inelastic at this point. [2]
- [5 marks]**

Q.22) Cortovia is experiencing a surge of cheap, imported steel, threatening the survival of its legacy domestic steel mills. The national steelworkers' union is demanding immediate tariffs and quotas, arguing that abrupt closures will devastate local communities.

Assess the economic arguments that Cortovia's government could use to justify imposing trade restrictions to protect its domestic steel industry.

[5 marks]

Q.23) Meridia, a developed economy, recently lifted its historical ban on aerospace exports and joined a global free trade network. AeroSat, a private Meridian manufacturer of low-Earth orbit (LEO) communication satellites, can now freely export its satellites and import specialized components like advanced optical sensors from abroad without tariffs.

Discuss the likely economic benefits that AeroSat and the broader Meridian economy will experience from this integration into the global aerospace market.

[4 marks]

Q.24)

- i) List down the merits of a central bank focusing exclusively on a simple inflation target. [3]
- ii) Discuss how the Taylor Rule offers an alternative monetary policy approach for setting interest rates in a modern economy. [3]

[6 marks]

Q.25) Explain the four key factors that determine a firm's degree of market power, and thus its ability to control the price of its product. [4 marks]

Q.26) Assess the crucial role central banks play in maintaining the overall functioning of an economy, particularly focusing on the inherent policy conflicts they must navigate when simultaneously attempting to control supply-side inflation and acting as the lender of last resort to a stressed banking sector.

[5 marks]

Q.27) In 2025, the Indian government awarded a 30-year exclusive concession to "VidyutGrid Corp," a newly formed private entity, to develop and manage a high-voltage direct current (HVDC) transmission network connecting renewable energy hubs in Rajasthan to major consumption centers in Maharashtra. The initial capital outlay for this infrastructure was

₹85,000 crores, and the cost of maintaining the network is relatively low and predictable. No other company is permitted to build competing transmission lines along this route.

VidyutGrid Corp currently operates without direct price regulation and charges state electricity boards a transmission fee per megawatt-hour (MWh).

- i) Explain, using the characteristics of the industry described above, why VidyutGrid Corp operates as a natural monopoly and how this acts as a barrier to entry. [2]
- ii) Discuss whether VidyutGrid Corp is likely to be allocatively and productively efficient in the long run, assuming its primary objective is profit maximization. [4]

[6 marks]

Q.28) To support the booming domestic real estate and infrastructure sectors, the government has recently announced a substantial reduction in the corporate profit tax rate specifically for domestic copper and electrical wiring manufacturers.

Discuss the likely impact of this corporate tax cut on the investment decisions and output of these electrical wiring firms and evaluate potential limitations of this policy in stimulating long-term growth in the sector.

[4 marks]

Q.29) Aura Grid is a modern renewable energy infrastructure firm planning to construct a series of offshore wind farms. This project requires massive, long-term capital investment and carries significant initial construction and technological risks. Conversely, domestic households in the economy have accumulated excess savings. These households are looking for a return on their savings but require short-term access to their funds for emergencies and possess limited expertise in evaluating complex energy infrastructure projects.

Discuss how the financial sector performs its core economic functions and reconcile the conflicting financial requirements of Aura Grid and the domestic households.

[5 marks]
